QC Portal

# Retrospective Study Module

Case intake, assignment and segmentation workflow

| | |
|---|---|
| Document | Functional and technical documentation of the work implemented so far |
| Project | QC Portal (QCPinkShieldAI), TANUH Foundation |
| Date | 28 October 2026 |
| Audience | Project team: admins, developers and reviewers of the QC Portal |
| State | Intake, dashboard, case viewer and assignment are built and tested. Radiologist flow, notes and the segmentation tool are planned (see section 10). |

# 1. Purpose and scope

The QC Portal reviews breast-cancer screening data in two studies. The **Prospective** study is data that keeps arriving from partner hospitals (clinic_00001 and so on) and is synchronised from the BCD portal database. The **Retrospective** study is data supplied manually, as a folder or zip, which is checked on upload and then segmented by Mammo Techs with Radiologist help.

This document covers the Retrospective module, which is the part built so far. The Prospective portal and a master dashboard combining both studies are planned and not started.

### What the Retrospective module does

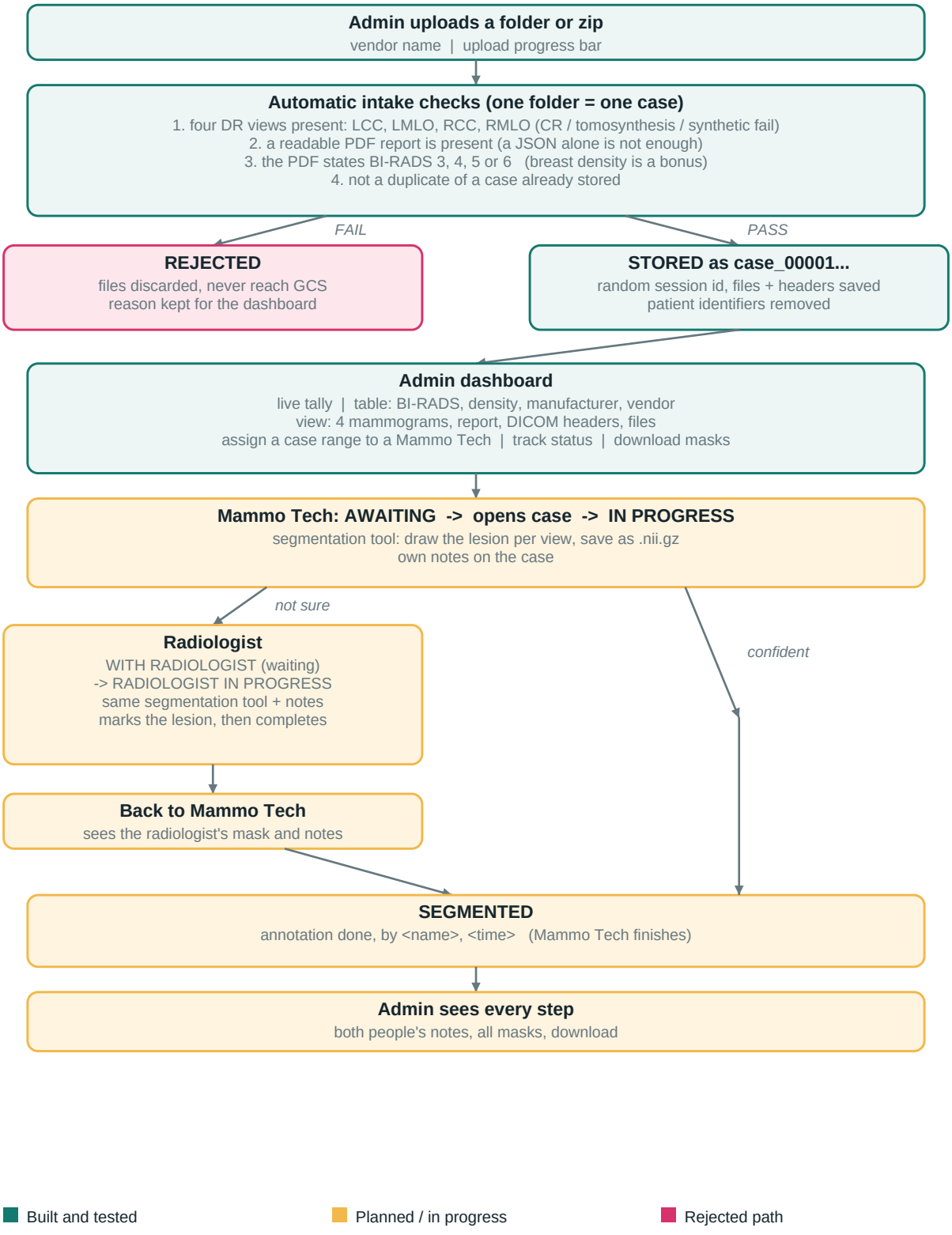
- Accepts a folder or zip upload and checks every case automatically; only qualifying cases are stored.
- Gives each passed case an ID (case_00001, ...) and a random unique session ID.
- Shows a live tally and a case table with BI-RADS, density, manufacturer, vendor, assignment, status and annotation.
- Lets the admin open a case read-only: four views, report, DICOM headers, files and segmentation progress.
- Lets the admin assign a range of cases to a Mammo Tech and follow progress; masks can be downloaded once segmented.

# 2. Roles

| Role | Does | Page |
|---|---|---|
| Admin | Uploads cases, sees tally and table, assigns ranges to Mammo Techs, opens cases read-only, downloads masks. | /qc/admin, tab Retrospective |
| Mammo Tech | Opens assigned cases, segments the lesion, saves masks, escalates to a Radiologist when unsure. (Backend ready; screen planned.) | /qc/mammo-tech |
| Radiologist | Receives escalated cases, marks the lesion with the same tool, writes clinical notes. (Planned.) | /qc/radiologist |

# 3. End-to-end flow

Green boxes are built and tested; amber boxes are planned or in progress.


Admin uploads a folder or zip
vendor name | upload progress bar
Automatic intake checks (one folder = one case)
1. four DR views present: LCC, LMLO, RCC, RMLO (CR / tomosynthesis / synthetic fail)
2. a readable PDF report is present (a JSON alone is not enough)
3. the PDF states BI-RADS 3, 4, 5 or 6 (breast density is a bonus)
4. not a duplicate of a case already stored
FAIL
PASS
REJECTED
files discarded, never reach GCS
reason kept for the dashboard
STORED as case_00001...
random session id, files + headers saved
patient identifiers removed
Admin dashboard
live tally | table: BI-RADS, density, manufacturer, vendor
view: 4 mammograms, report, DICOM headers, files
assign a case range to a Mammo Tech | track status | download masks
Mammo Tech: AWAITING -> opens case -> IN PROGRESS
segmentation tool: draw the lesion per view, save as .nii.gz
own notes on the case
not sure
confident
Radiologist
WITH RADIOLOGIST (waiting)
-> RADIOLOGIST IN PROGRESS
same segmentation tool + notes
marks the lesion, then completes
Back to Mammo Tech
sees the radiologist's mask and notes
SEGMENTED
annotation done, by <name>, <time> (Mammo Tech finishes)
Admin sees every step
both people's notes, all masks, download
Built and tested
Planned / in progress
Rejected path

# 4. Intake rules

A case is one folder. A folder may hold the DICOMs in a sub-folder (for example DICOM/) next to the report files. All checks run automatically after upload. If any check fails, the whole case is rejected, its files are discarded, and only the reasons are kept.

| Check | Passes when | Otherwise |
|---|---|---|
| Four views | Exactly one DICOM each for LCC, LMLO, RCC, RMLO. View is read from the DICOM (laterality and view position), falling back to folder or file names. | Rejected: missing view, duplicate view, or view not determined |
| Readable DICOM | File opens as DICOM and has image dimensions. | Rejected: unreadable DICOM |
| DR (digital mammogram) | Modality is MG, SOP class is Digital Mammography X-Ray (presentation or processing), and the series description, protocol or image type does not mention tomosynthesis, DBT or synthetic 2D. | Rejected: not DR (CR, tomosynthesis, synthetic) |
| PDF report | At least one PDF with extractable text. A JSON alone is not enough. | Rejected: PDF missing or unreadable |
| BI-RADS | The PDF states BI-RADS, and it is 3, 4, 5 or 6 (4A, 4B, 4C count as 4). | Rejected: BI-RADS missing, or not allowed |
| Breast density | Optional. Read from the PDF, or from a JSON if the PDF has none. | Not a reason to reject |
| JSON report | Optional. If present and valid it is stored with the case. An invalid JSON is ignored. | Not a reason to reject |
| Duplicate | The four images are not byte-identical to a case already stored. | Rejected: duplicate of case_xxxxx |

### Privacy

Patient name, birth date, address, telephone and physician names are left out of the DICOM header cache that admins see. The original files are stored unchanged in the bucket, so de-identification of the source files remains the data provider's responsibility.

### Report reading

PDF text is extracted with pdftotext. The BI-RADS value is read from the report's impression section; the reference legend that lists every category is ignored. Scanned PDFs without a text layer cannot be read and are rejected; OCR is not implemented.

# 5. Case lifecycle and statuses

| Status | Meaning | Set by |
|---|---|---|
| Unassigned | Passed intake; no Mammo Tech yet. | Intake |
| Awaiting | Assigned to a Mammo Tech who has not opened it. | Admin assigns a range |
| In progress | Mammo Tech is segmenting. | Mammo Tech opens the case |
| With radiologist | Mammo Tech is unsure about the lesion and escalated (a note is required). Waiting for the Radiologist. | Mammo Tech escalates |
| Radiologist in progress | Radiologist has opened the case. (Planned.) | Radiologist |
| Segmented | Mammo Tech finished; annotation is done, by whom and when. | Mammo Tech |

### Assumptions made (please confirm)

- After the Radiologist completes, the case returns to the Mammo Tech, who sees the Radiologist's mask and notes and finalises the segmentation.
- One mask file per breast view (four possible per case), not one per case. The mask has the same width and height as the DICOM pixel grid.
- When a case is escalated, the Radiologist with the fewest open retrospective cases is chosen, instead of a random pick.
- Admins only read; they do not edit masks or notes.

# 6. Data model

## Databases

| Database | Role | Access from QC app |
|---|---|---|
| qc_bcd_portal | QC application data: users, roles, hospitals, assessments, assignments. The new qc_retro_* tables live here. | Read and write |
| qc_bcd_questionnaire | QC copy of the questionnaire data. | Read and write |
| bcd_application2, bcd_questionnaire | Live BCD prospective source. Synchronised into QC by database/cloud/qc_sync_catchup.sql. | Read-only; never written |
| qc_retrospective | Existing inventory from earlier ingestion (cases, files, upload batches; 1,091 cases). Separate schema. | Not used by the new flow; not imported |

## New tables (all in qc_bcd_portal)

| Table | Holds |
|---|---|
| `qc_retro_batches` | One row per upload: vendor, status, phase, counts (received, checked, passed, rejected). Drives the live tally. |
| `qc_retro_cases` | Only cases that passed. case code, random session id, content hash (blocks duplicates), original name, modality, BI-RADS, density, manufacturer, vendor. |
| `qc_retro_files` | Each stored file: kind (view or report), view, filename, folder path inside the case, GCS path, size, SHA-256, DICOM header cache. |
| `qc_retro_rejections` | Audit row for each rejected case with the list of reasons. No files kept. |
| `qc_retro_assignments` | One row per assigned case: Mammo Tech, status, timestamps, escalation note, Radiologist (from migration 3). |
| `qc_retro_masks` | Segmentation masks (.nii.gz): version, view, author role, author, size, timestamp. |
| `qc_retro_notes` | Notes thread on a case: Mammo Tech doubts, Radiologist clinical notes and guidance. (migration 3) |

## Migrations (apply in this order, to qc_bcd_portal only)

| File | What it does |
|---|---|
| `20261028_add_retro_intake.sql` | Creates batches, cases, files and rejections tables. |
| `20261029_add_retro_assignments.sql` | Creates assignments and masks tables. |
| `20261030_retro_radiologist_notes_masks.sql` | Adds source path to files; Radiologist columns and a wider status column on assignments; view and role on masks; creates notes table. Not idempotent: run once. |

None of the migrations touch an existing table. The application does not apply migrations at startup; they are run by hand.

# 7. Storage layout

Files that pass are written to the bucket configured as GCP_STORAGE_BUCKET (breast-cancer-image-dataset), under a configurable prefix (default retrospective/):

```
retrospective/case_00001/<VIEW>/<file>.dcm
retrospective/case_00001/report/<file>.pdf and .json
retrospective/case_00001/segmentation/<file>.nii.gz
```

Uploads are staged on the server disk while checks run, then deleted whether the batch succeeds or fails. Rejected cases are never copied to the bucket.

Existing folders such as retrospective/RETRO_000003/ from the earlier ingestion are not touched.

# 8. API reference (current)

All paths are under /api/v1/qc/retro and need a Bearer token. Admin endpoints require the Admin role; Mammo Tech endpoints require Mammo Tech.

| Method and path | Role | Purpose |
|---|---|---|
| `POST /batches` | Admin | Start an upload; returns batch id. Form: source_name, vendor. |
| `POST /batches/{id}/files` | Admin | Send one chunk of files (files + paths). Called repeatedly. |
| `POST /batches/{id}/finalize` | Admin | Stop receiving and start the checks in the background. |
| `GET /batches/{id}` | Admin | Progress: phase, checked / total, passed, rejected, error. |
| `GET /batches/{id}/rejections` | Admin | Rejected cases with reasons. |
| `GET /tally` | Admin | Live totals, per-status counts, BI-RADS and density breakdown, unassigned range. |
| `GET /cases` | Admin | Case list. Filters: q, birads, density, status, mammo_tech_id; paged. |
| `GET /cases/{id}` | Admin | Case detail: files, DICOM headers, assignment, annotation, masks. |
| `GET /files/{id}/image` | Admin | Mammogram rendered as PNG (server-side windowing). |
| `GET /files/{id}/content` | Admin | Raw report (PDF or JSON). |
| `GET /mammo-techs` | Admin | Active Mammo Techs with workload counts. |
| `POST /assign` | Admin | Assign every unassigned case in a number range to one Mammo Tech. Existing assignments are kept. |
| `POST /unassign` | Admin | Take back cases that are still Awaiting. |
| `GET /masks/{id}/download` | Admin | Download a mask (.nii.gz). |
| `GET /my-cases` | Mammo Tech | Cases assigned to me. |
| `POST /my-cases/{id}/start` | Mammo Tech | Awaiting to In progress. |
| `POST /my-cases/{id}/escalate` | Mammo Tech | Send to Radiologist; a note is required. |
| `POST /my-cases/{id}/mask` | Mammo Tech | Upload a .nii.gz; header validated; re-upload makes a new version. Will become per-view. |

Planned: Radiologist endpoints (my cases, start, complete), notes on a case, per-view mask upload with a check that the mask size matches the DICOM, a zip download of a case's masks, and read access to a case for its assigned Mammo Tech and Radiologist.

# 9. Using the admin page

Log in as Admin and open the Retrospective tab in the sidebar.

- **Banner and stat cards:** cases received, checked, passed, rejected; a pass-rate donut; bars for BI-RADS and breast density of passed cases. The tally refreshes every few seconds.
- **Upload:** enter the vendor or supplying organisation, then drop a .zip, or choose a folder or zip. A three-step tracker (Upload, Checks, Stored) and two progress bars show upload and checking progress, with live passed and rejected counts. When finished, each rejected case is listed with its reasons.
- **Assign to Mammo Tech:** pick a Mammo Tech and a From and To case number (for example 1 to 20), or use the button to fill the whole unassigned range. Cases already assigned are left unchanged and reported. A card per Mammo Tech shows a progress bar by status.
- **Passed cases table:** case, original name, modality, BI-RADS, density, manufacturer, vendor, assigned to, status, and annotation (done, by whom, when). Filter by BI-RADS or status, or search. A Mask button downloads the segmentation once a case is segmented.
- **View a case:** opens a read-only window. Tabs: Mammograms (four views in the standard layout with zoom, pan, brightness, contrast, invert), Report (PDF or JSON), DICOM headers (searchable per view), Files (names and sizes), Segmentation (timeline, masks and downloads).

# 10. Build status

| Item | State |
|---|---|
| Upload with progress bar, automatic checks, rejection reasons, live tally | Built, tested |
| New rule: PDF mandatory with BI-RADS; JSON optional; density bonus; folder path stored | Built, tested (18 intake tests) |
| Case table, charts, case viewer (views, report, headers, files) | Built; checked against sample data in a sandbox, not yet reviewed in your browser on real data |
| Assignment by case range, Mammo Tech progress, mask download | Built, tested |
| Mammo Tech escalation (backend) | Built, tested; no screen yet |
| Migration 20261030 (Radiologist, notes, per-view masks, source paths) | Written, not applied |
| Radiologist flow, case notes, per-view masks | Not built yet |
| Segmentation tool (brush, eraser, polygon, undo, overlay, .nii.gz export) for Mammo Tech and Radiologist | Not built yet |
| Admin view of the case folder and the Radiologist's progress and notes | Not built yet |
| Prospective portal (hospitals, CR/DR filter, DICOM and report viewing) | Not started |
| Master dashboard combining both studies | Not started |

## Known limitations

- The checks depend on DICOM tags being filled correctly; a wrong SOP class or modality causes a rejection.
- Scanned PDFs are rejected because OCR is not implemented.
- The 1,091 existing cases in the qc_retrospective schema are not imported. Decide whether they should enter this flow.
- Rendering of real compressed DICOMs depends on the decoder libraries in requirements and has only been tried with sample images.

- No email is sent when cases are assigned or escalated yet.
- The reverse proxy (nginx) must allow large request bodies for upload chunks (about 50 MB or more); this has not been changed.
- The backend test suite had 82 failing tests before this work started, and a test file that cannot be imported; these are unrelated and unchanged.

# 11. Setup and running

## Production

- Apply the three migrations, in order, to qc_bcd_portal (take a backup first).
- Make sure GCP_STORAGE_BUCKET is set and the service account can create and read objects under the retrospective/ prefix.
- Optional settings: RETRO_GCS_PREFIX (default retrospective/) and RETRO_STAGING_DIR (default: system temp directory, needs room for the largest upload).
- Deploy backend and frontend together. The Retrospective tab appears for Admin users.

## Local sandbox (no real database or bucket)

- A throwaway SQLite database and a local folder stand in for Cloud SQL and the bucket. Nothing is sent to GCP.
- Backend on port 8002 and frontend on port 3001, so existing servers on 8000, 8001 and 3000 are left alone.
- Demo data is synthetic and clearly labelled; it disappears when the sandbox is cleaned up.

## Tests

```
python -m pytest backend/tests/test_retro_intake.py backend/tests/test_retro_assign.py
```

The assignment tests will be updated when masks become per-view.

## Files added

| Area | Files |
|---|---|
| Backend | api/retro.py, api/retro_assign.py, services/retro_intake.py, services/retro_workflow.py, models/retro.py, main.py (router registration) |
| Frontend | components/RetroIntake.js and .css, components/RetroCaseViewer.js and .css, pages/AdminPage.js (Retrospective tab) |
| Database | migrations 20261028, 20261029, 20261030 |
| Tests | tests/test_retro_intake.py, tests/test_retro_assign.py |